電子回路論　坂本宏　room 303

NN scphys.exp.1cairo

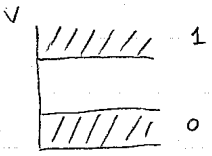
No. 1

Date 94・10・4

論理回路

二値化

0	1
NO	YES
LOW	HIGH



ブール代数

$0+0=0$, $0+1=1$ 論理和

$1+0=1$, $1+1=1$

$0\cdot0=0$, $0\cdot1=0$ 論理積

$1\cdot0=0$, $1\cdot1=1$

$\bar{1}=0$, $\bar{0}=1$ 論理否定

$\overline{(A\cdot B)}=\bar{A}+\bar{B}$ ドモルガンの定理

$\overline{(A+B)}=\bar{A}\cdot\bar{B}$

$\overline{(A+B+C)}=\bar{A}+\bar{B}+\bar{C}$

$A\cdot B=B\cdot A$, $A+B=B+A$ 交換則

$A\cdot1=A$, $A\cdot0=0$

$A+1=1$, $A+0=A$ 吸収則

$A\cdot(B\cdot C)=(A\cdot B)\cdot C$

$A+(B+C)=(A+B)+C$ 結合則

$A\cdot(B+C)=A\cdot B+A\cdot C$

$A+(B\cdot C)=(A+B)(A+C)$ 分配則

$=A\cdot A+A\cdot C+B\cdot A+B\cdot C$

$=A+A\cdot C+A\cdot B+B\cdot C$

$=A(1+C+B)+B\cdot C$

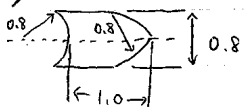
$=A+B\cdot C$

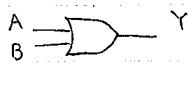
A \ C	0		1	
	B 0	1	1	0
0	1	1	1	1
1	1	1	0	1

カルノー図

論理記号　MIL記号

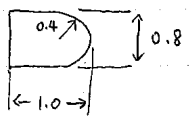
論理和

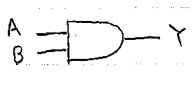




$Y=A+B$ OR

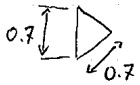
論理積





$Y=A\cdot B$ AND

通過



$Y=A$ BUF

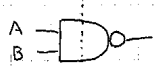
論理否定



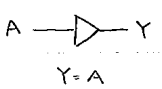
NOT

ex) 正論理：負論理

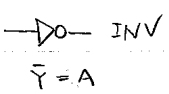
NAND



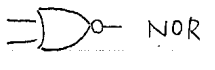
$Y=\overline{A\cdot B}$



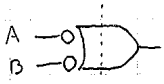
$Y=A$



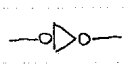
$\bar{Y}=A$



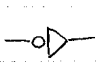
INV-OR



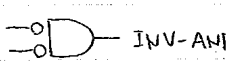
$Y=\bar{A}+\bar{B}$



$\bar{Y}=\bar{A}$

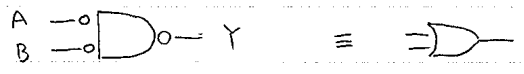


$Y=\bar{A}$



No. 2

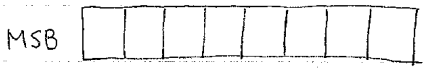
Date 94 · 10 · 4



$\bar{Y} = \bar{A} \cdot \bar{B}$ $\qquad Y = A + B$



二進法 0 1 10 11 100 101 110 111 1000



Least Significant Bit

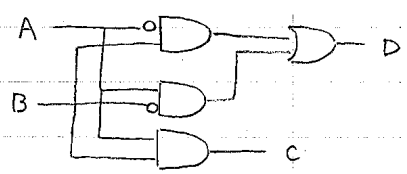
2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

256 128 64 32 16 8 2 1

A	B	A+B
0	0	00
0	1	01
1	0	01
1	1	11 ← 10?
		C D

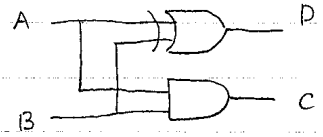
$C = A \cdot B$

$D = \bar{A} \cdot B + A \cdot \bar{B}$

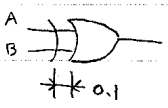


Carry

A \ B	0	1
0	0	1
1	1	0



排他論理和 XOR



$Y = A \oplus B$

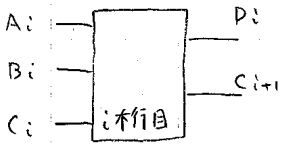
A_0	B_1 B_0	A_1 A_0			$D_1 D_0$
A_1	0 0	0 0	0	10	00 10
	0 0	0 1	1	10	01 11
B_0	0 0	1 0	10	10	10 100
	0 0	1 1	11	10	11 101
B_1	0 1	0 0	1	11	00 11
	0 1	0 1	10	11	01 100
	0 1	1 0	11	11	10 101
	0 1	1 1	100	11	11 110

$D_0 = A_0 \oplus B_0$

$D_1 = f(A_1, B_1, C_1)$

$D_2 = f(A_2, B_2, C_2)$

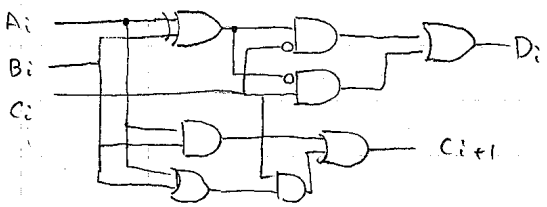
加算器



Ci:	0		1	
Bi:	0	1	1	0
Ai: 0	0	1	0	1
1	1	0	1	0

$D_i = \bar{C}_i \cdot (A \oplus B) + C_i \overline{(A \oplus B)}$

$C_{i+1} = A_i B_i + C_i (B_i + A_i)$

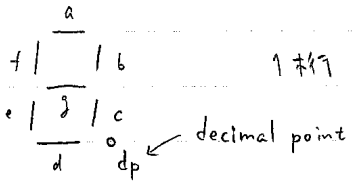


(1) 4桁 + 4桁 の加算器

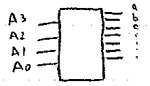
(2) 乗算器のアイデア

▷ Oct 11

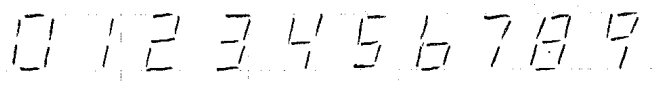
Sequential Logic
順序回路



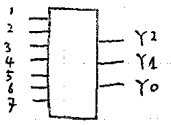
1桁 4 bit



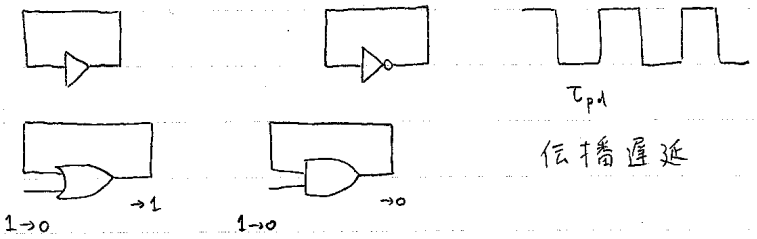
デコーダー



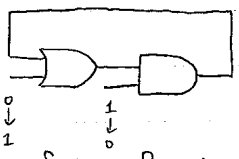
a = (A==0) OR (A==2) OR (A==3) OR (A==5) OR (A==7) OR (A==8) OR (A==9)



ENCODE

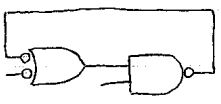


伝播遅延

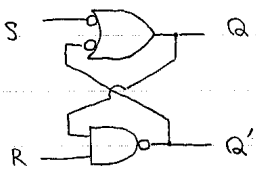


Set Reset

FlipFlop FF



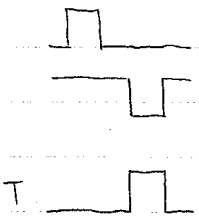
Set Reset

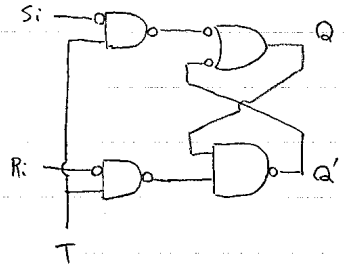


RS-FF

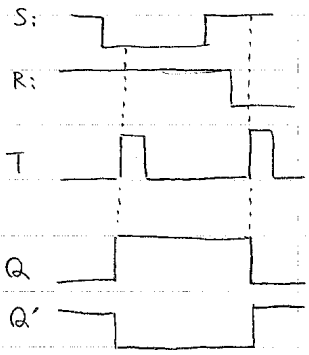
No. 4

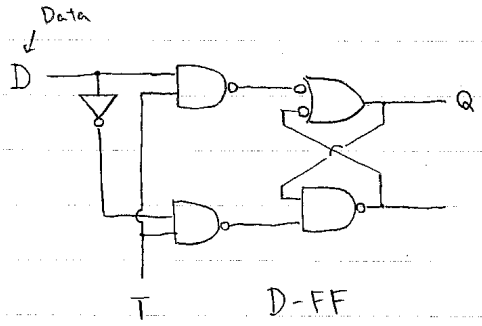
Date . .



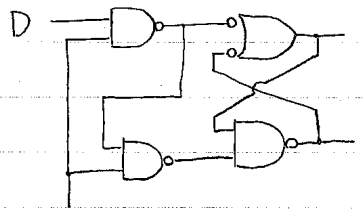


RST-FF

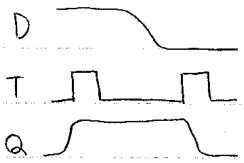




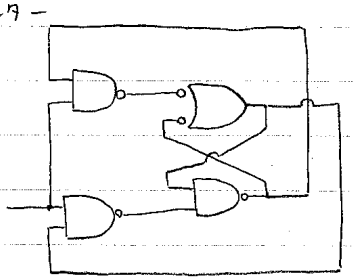
D-FF

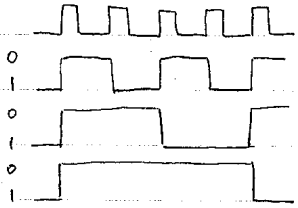


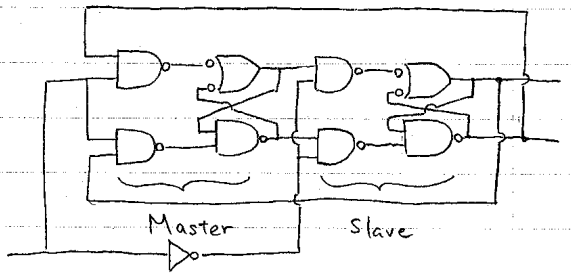
(D·T)·T

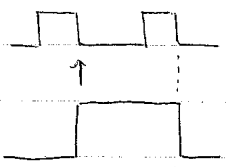


カウンター

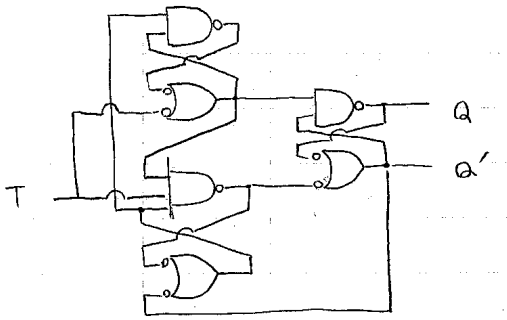




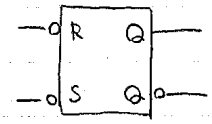


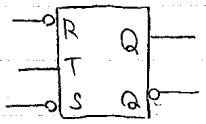


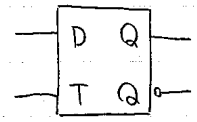
Master Slave FF

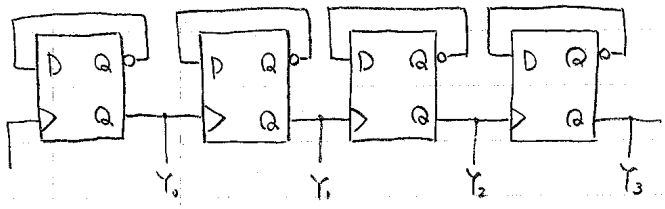


Edge Trigger

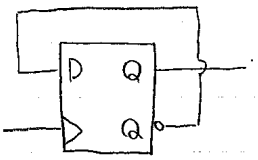








4bit カウンター

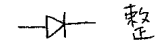


(1) Edge Trigger のカウンターに Clear 線をつなぐ (全体を ゼロにする)
〃 初期状態にする

▷ Oct 18

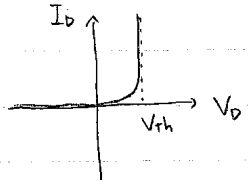
TTL (トランジスタ トランジスタ ロジック) c.f. DTL

半導体

ダイオード  整流

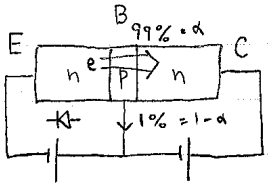
順方向電圧降下 Si の場合 0.7V

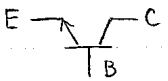
絶対最大定格



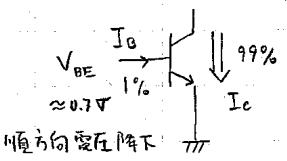


トランジスタ





$\frac{I_C}{I_B} = \frac{\alpha}{1-\alpha} = \beta \sim 100 - 1000$



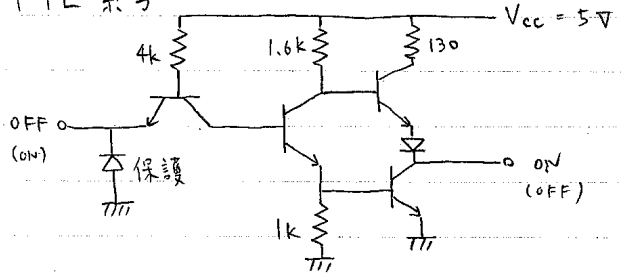
エミッタ接地

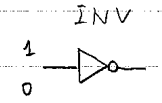
直流電流増幅率

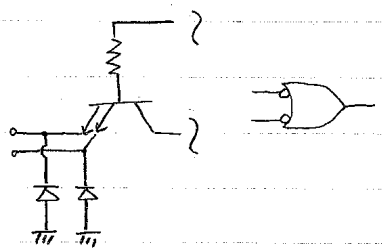
No. 6

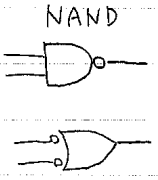
Date 94・10・18

TTL 素子

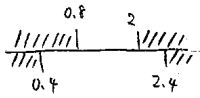








	Low	High
V_{in}	< 0.8V	2V <
V_{out}	< 0.4V	2.4V <
I_{in}	-1.6mA	40μA
I_{out}	16mA	-400μA

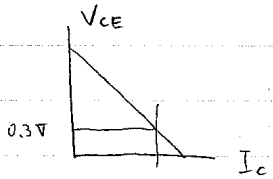


(N-TTL) t_{pd} = 22nSec

normal

propagation delay

伝播遅延



0.3V 順方向電圧降下



ショットキーバリアダイオード

金属とSiの接合面

LS-TTL

Low power

	Low	High
V_{in}	< 0.8V	2.0V <
V_{out}	< 0.5V	2.7V <
I_{in}	-0.36mA	20μA
I_{out}	8mA	-400μA

t_{pd} = 15nSec

ALS-TTL

Advanced

	- 0.1mA	20μA
	8mA	-400μA

AS-TTL

ショットキークランプダイオード

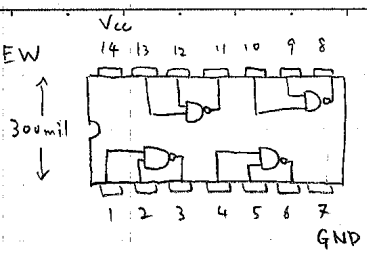
S-TTL τ_{pd} = 4.5nSec

Low	High
-2mA	50mA
20mA	-1000mA

No. 7

Date 94 · 10 · 18

Top VIEW



V_{cc} 4.75 V ~ 5.25 V

SN7400

SN74LS00

課題 TTL 等価回路

NOR

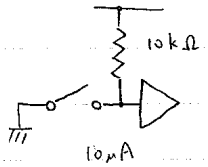
BUFFER

1 mil = 1/1000 inch

▷ Oct. 25

TTLと他の電子部品との接続

スイッチ



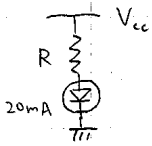
発光ダイオード LED (light Emitted Diode)

$10k\Omega \times 10\mu A = 100mW$

$\frac{5V}{10k\Omega} = 0.5mA$

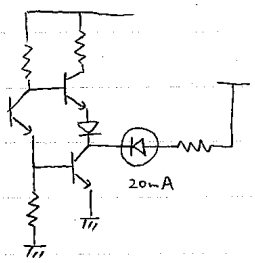
順方向電圧降下 ≒ 可視光のエネルギー

赤 1.6 eV

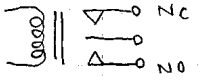


$I_R = V_{cc} - V_{LED}$

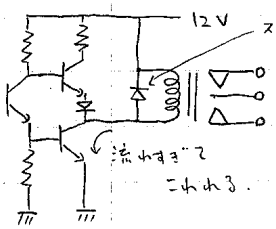
$20mA \cdot R = 5V - 1.6V$ $R = 170\Omega$

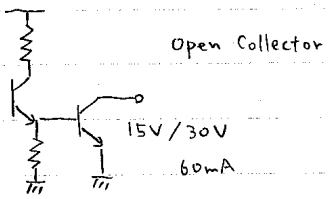


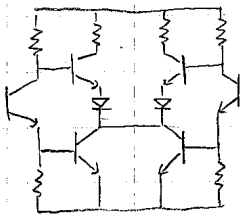
リレー

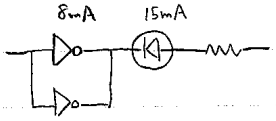


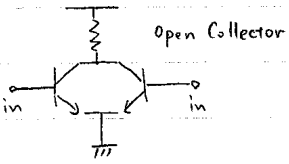
5V 20mA

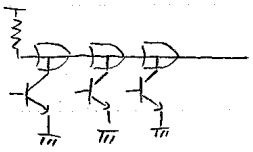










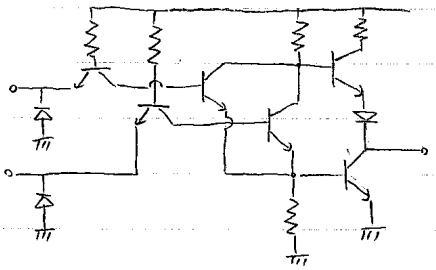


No. 8

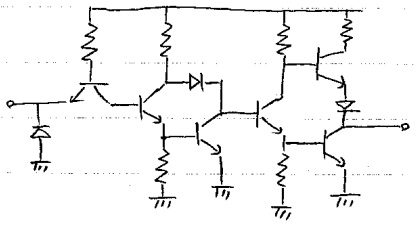
Date 94. 10. 25

課題の解答

NOR



BUFFER



TTL回路の電源

5V 安定化 4.75 - 5.25 V

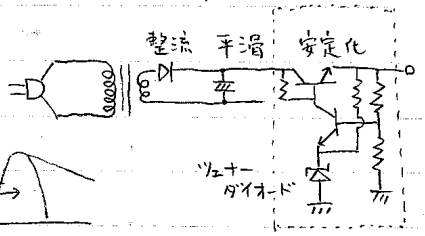
6V 安定化 → 6V —▷— 5.3V

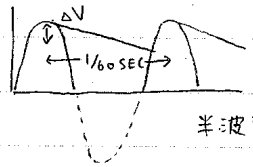
CAMAC

12V 安定化 —▷—···—▷— 1A

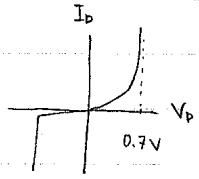
定電圧電源 10ヶ

100V AC

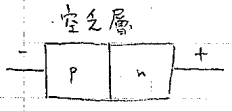




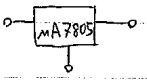
半波整流



降服

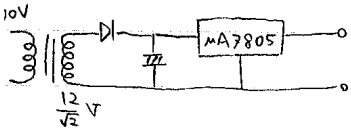


雪崩をおこす

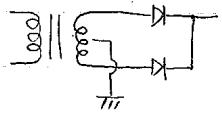


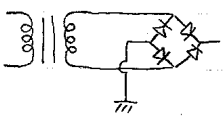
三端子レギュレーター

8 05
+ 5V



+7 ~ 12V 入力 +5V 出力 1A





ブリッジ回路

$C\Delta V = \Delta Q = I\Delta T$

(12-7) 1A、1/120 SEC

$C = \frac{1}{600}$ F

電源を設計

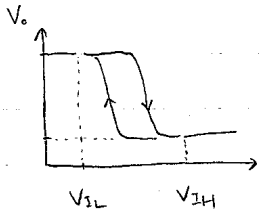
- 入力電圧 80 ~ 120V
- ブリッジ整流
- 出力 5V 1A
- μA7805

Small Median Large
SSI - MSI - LSI

トーテムポール ?

TTL 74 **
LS **
S **
ALS**
AS**

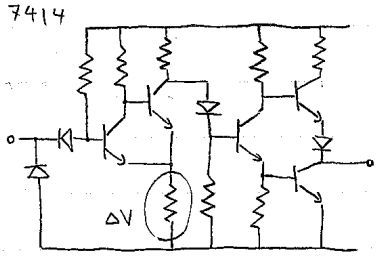
VIH IIH
VIL IIL
VOH IOH
VOL IOL

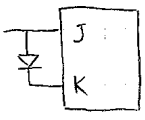


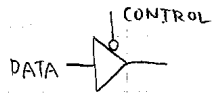


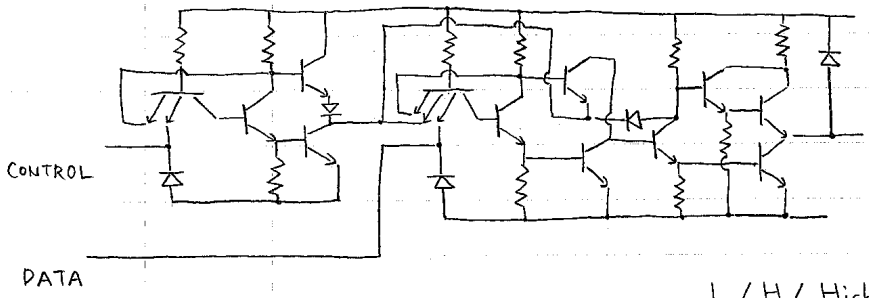
シュミット・トリガー

7414



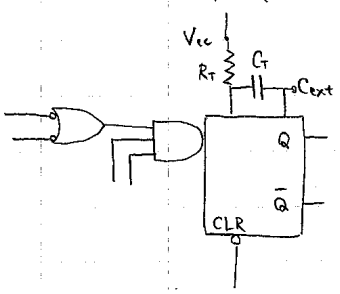


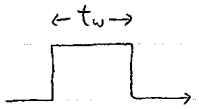




L / H / High Z Three state

Rext / Cext



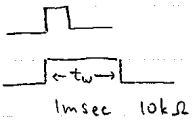


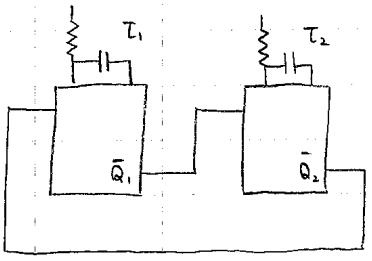
$t_w = 0.45 R_T C_T$

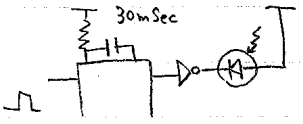
(LS)

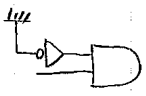
$5k\Omega < R_T < 260k\Omega$

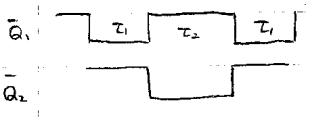
$1msec = 0.45 \times 10k\Omega \cdot C_T$ $C_T = \frac{1}{5}\mu F$





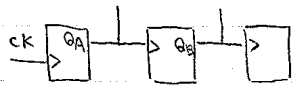


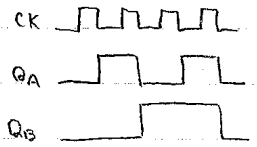




No.

Date





ストップウォッチをつくる

1/100 sec ~ 10000 sec　6桁

START　STOP　CLEAR　スイッチ

